

Terrain-Conditioned Deep Neural Networks for Next-Day High-Flow Prediction in Sri Lanka

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Abstract—Deep neural networks can combine antecedent rainfall, river discharge and catchment characteristics for flood-related prediction, but their usefulness depends on both the representation learned and the validity of the evaluation. This study examines MMF-Net, a terrain-conditioned temporal transformer for next-day high-flow prediction at 51 river-basin locations in Sri Lanka. The network encodes 14-day sequences of 33 dynamic variables using periodic numerical embeddings, temporal self-attention and cross-feature attention, with nine static attributes supplied through feature-wise linear modulation. An optional Sentinel-1 branch supports an exploratory assessment of radar fusion. The primary target is exceedance of a training-period discharge percentile in GloFAS-derived data; it is a hydrological proxy rather than an observed inundation label. In the recorded temporal experiments, replacing linear embeddings with periodic embeddings increases test average precision from 0.6083 to 0.7605. The terrain-conditioned five-seed ensemble trained with binary cross-entropy reaches 0.8355, with a Brier score of 0.0080 and expected calibration error of 0.0016. These results support further investigation of numerical representations and static conditioning in neural flood models. Single-seed ablations, incomplete radar controls and limited agreement between proxy labels and documented floods constrain interpretation. We identify the observations and evaluation steps required to assess the model as a flood-warning decision-support component.

Index Terms—deep neural networks, temporal transformer, numerical embeddings, terrain conditioning, flood prediction, Sri Lanka

I. INTRODUCTION

Predicting high river flow requires information about both recent weather and the state of the catchment. Similar rainfall totals can produce different responses depending on antecedent wetness, existing discharge and local conditions. A neural forecasting model must therefore represent interactions among variables as well as their evolution through time. In Sri Lanka, a dataset spanning contrasting climatic zones provides an opportunity to study these representations across multiple river-basin locations.

Recurrent neural networks have demonstrated the value of learning rainfall–runoff relationships from hydrological sequences [1]. More recent large-scale work has extended neural flood forecasting to ungauged watersheds [2]. These results motivate regional studies, but the distinction between the prediction target and the intended application remains essential. A network trained to reproduce a discharge reanalysis is evaluated against that reanalysis. Its performance cannot be

interpreted directly as accuracy against measured water levels, inundation extent or damage.

This paper investigates MMF-Net, a deep network that combines a temporal transformer with periodic numerical embeddings, cross-feature attention and static conditioning. The research emphasis is on neural representation and fusion: how continuous hydrometeorological inputs are encoded, how static attributes modify the learned state, and whether an additional radar branch provides useful information. Conventional reference models provide context for the difficulty of the target; the principal experiments concern the DNN and its component ablations.

Three questions guide the analysis: (1) how do numerical embeddings, cross-feature attention and terrain conditioning affect the recorded neural prediction scores; (2) how do the training objective and ensemble calibration affect ranking and probability quality; and (3) what evidence is available for radar fusion and practical warning use? The contribution is an application study of established neural components in Sri Lanka, with explicit experimental controls and limitations.

II. RELATED WORK

A. Neural hydrological forecasting

Following neural rainfall–runoff modelling [1], large-scale studies have demonstrated prediction of extreme river flows in ungauged watersheds [2]. Here the task is daily classification at fixed Sri Lankan locations using reanalysis supervision, which differs from operational global forecasting in its scale, target and evaluation.

Local research also motivates neural temporal models. Saubhagya *et al.* combine a bidirectional LSTM with time-series regression for water-level forecasting in the Ratnapura area of the Kalu basin [3]. Differences in target, horizon and data prevent direct comparison with the present multi-location exceedance task.

B. Representation, conditioning and spatial structure

Transformers use self-attention to represent relationships among sequence elements [4]. For continuous inputs, Gorishniy *et al.* show that feature embeddings can improve deep models across several backbones [5]. We apply their periodic embedding approach before temporal encoding. Feature-wise linear modulation (FiLM) conditions a learned representation

on auxiliary attributes [6]; here these describe terrain, climatic zone and network position.

River graphs offer another way to incorporate hydrological structure, although their benefit must be established against a suitable per-location control. Kirschstein and Sun found that their neural forecasting models did not benefit from the supplied river topology [7]. This motivates a controlled question rather than a presumption that adding edges improves forecasting. MMF-Net is graph-free: weights are shared across locations, but predictions do not exchange messages between them. Earlier graph-network experiments are discussed separately because their encoder and data extent differ from those of the main DNN.

C. Probability quality and evaluation

Focal loss was introduced to address the dominance of easy examples in dense object detection [8]; its suitability for hydrological classification is an empirical question. Deep ensembles and temperature scaling provide established approaches to probability estimation and post-hoc calibration [9], [10]. Under rare positive events, precision-recall analysis complements threshold-based verification [11]. Temporally structured observations also require an evaluation design that respects their dependence [12]. These ideas inform our interpretation of the DNN results and their practical limits.

III. STUDY DATA AND PREDICTION TASK

A. Spatial coverage and input provenance

The node inventory contains 51 locations in 16 basin or area groups, including the Colombo metropolitan group. It assigns 31 nodes to the wet zone, 13 to the dry zone and seven to the intermediate zone. Fig. 1 shows their coordinates and the 35 curated directed flow links. These are representative modelling locations; the experiments do not use measured discharge from 51 gauges. The graph files also contain 204 spatial links for the separate graph-based models. Lines in the map indicate connectivity and do not trace surveyed river channels.

NASA POWER supplies base daily meteorological and soil-wetness variables [13]. The feature builder preferentially uses cached Open-Meteo Archive precipitation where available, falling back to POWER precipitation. The archive request does not pin a particular reanalysis model version. Discharge is obtained through the Open-Meteo Flood API from the GloFAS product family [14], [15]. Elevation in the supplied node table comes from POWER response metadata. Product versions and the proportions of precipitation sources need to be retained in a data manifest to reproduce the exact historical panel.

The 33 dynamic inputs comprise 11 rainfall and antecedent-rainfall variables, seven temperature, humidity, wind and radiation variables, six soil-wetness variables, and nine discharge variables. Discharge features include recent changes, rolling means, seasonal anomalies and the empirical percentile. The nine static inputs comprise elevation, log mean-training-discharge as a drainage proxy, three climatic-zone indicators and four network-position indicators. The drainage proxy is not a measured catchment area.

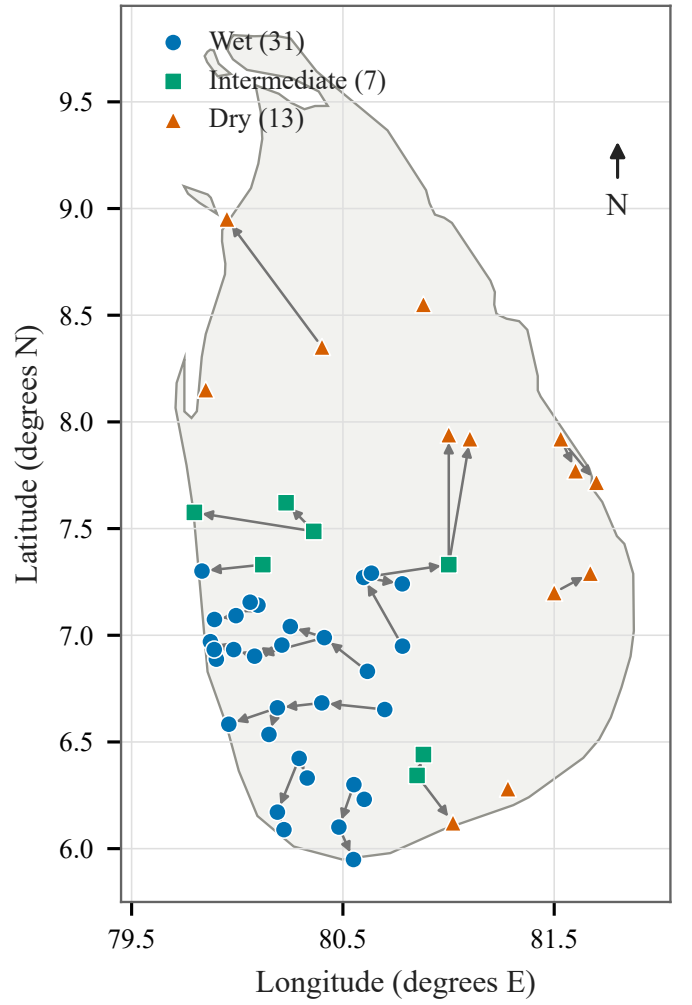


Fig. 1. Study locations and curated upstream-to-downstream links. Colours and marker shapes identify the inventory’s climatic-zone categories. Country outline: Natural Earth. Hydrological basin boundaries are not shown.

Selected heavy-tailed dynamic variables undergo a signed $\log(1 + |x|)$ transformation followed by standardisation. Dynamic statistics are fitted on the temporal training subset. Discharge percentile references and climatologies are fitted through 2017. Static continuous attributes are standardised across the node inventory. Missing normalised dynamic values are replaced with zero, corresponding to the training mean; the main tabular stream has no separate missingness mask.

B. Targets and episode definition

Let $Q_{n,t}$ denote reanalysis discharge at location n on day t . The high-flow state and primary next-day target are

$$\tau_n = \text{quantile}_{0.98}\{Q_{n,t} : t \leq 2017-12-31\}, \quad (1)$$

$$s_{n,t} = \mathbf{1}[Q_{n,t} \geq \tau_n], \quad y_{n,t}^{(1)} = s_{n,t+1}. \quad (2)$$

This percentile of daily discharge is a relative high-flow threshold, not a calibrated return-period estimate or an official warning-stage threshold. It does not establish whether water leaves the channel or affects settlements.

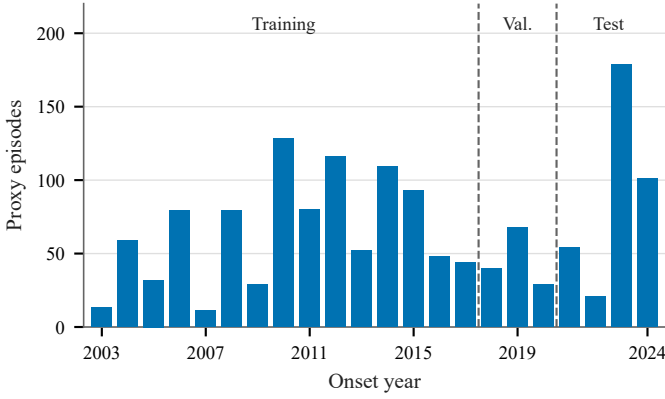


Fig. 2. Annual counts of node-level proxy episodes with onset in 2003–2024, computed from the supplied catalogue (1,464 entries). Divisions indicate the temporal training, validation and test periods. These are catalogue counts, not the episodes eligible for a particular model’s evaluation.

Auxiliary classification targets indicate any exceedance in the next two and three days, $y_{n,t}^{(h)} = \max_{k=1,\dots,h} s_{n,t+k}$, and next-day onset, $y_{n,t}^{(o)} = (1 - s_{n,t})s_{n,t+1}$. Two regression heads predict the next-day discharge percentile and the maximum discharge z-score over the next three days. All reported main-model scores concern $y^{(1)}$, not the onset head.

Episodes are constructed separately at each node by grouping positive state days whose successive dates are at most two days apart. A one-day non-exceedance gap may therefore lie inside an episode span. These are node-level proxy episodes: a single storm can produce several episodes at different nodes. They are not independent national disasters.

C. Coverage and data-version scope

The main DNN configuration truncates inputs after 31 December 2024. The supplied episode catalogue contains 1,469 entries through 2025, of which 1,464 start in 2003–2024. Fig. 2 summarises the retained catalogue without interpreting its annual variation as a trend in observed disasters.

The earlier validation report lists 409,350 valid node-days, including 75,555 test records, but covers a panel extending into 2025. Those totals cannot describe the truncated DNN experiment without a new panel audit. We report the protocol dates in Table I and retain historical performance values as recorded. Exact post-filter sample counts must be recovered from run artifacts before submission; the full panel and saved predictions are absent from the current paper checkout.

IV. DEEP NEURAL NETWORK METHOD

A. Numerical and temporal representation

For a prediction issued after day t , the input is $\mathbf{X}_{n,t} \in \mathbb{R}^{14 \times 33}$, ending on that day. Nodes are folded into the batch dimension, so the same network processes every node’s history independently. For scalar feature j , the periodic–linear–ReLU (PLR) representation is

$$\mathbf{e}_j(x) = \text{ReLU}(\mathbf{W}_j[\sin(2\pi\mathbf{c}_jx) \parallel \cos(2\pi\mathbf{c}_jx)] + \mathbf{b}_j), \quad (3)$$

where $\mathbf{c}_j \in \mathbb{R}^8$ contains trainable frequencies and $\mathbf{e}_j \in \mathbb{R}^8$. This supplies a flexible nonlinear representation before features are mixed; it does not imply that an ordinary deep network cannot learn nonlinear boundaries.

Each day’s feature embeddings are concatenated and projected to a 128-dimensional token. Learned position embeddings and a classification token are passed through three pre-layer-normalised transformer blocks, each with four attention heads, a feed-forward width of 256 and dropout of 0.2. The final classification-token state is the temporal representation $\mathbf{h}^T$.

B. Cross-feature attention and static conditioning

The feature branch uses the same embeddings, averaged over the 14-day window. Each feature is projected to a 128-dimensional token with a learned feature identity. A one-layer transformer pools these tokens to produce $\mathbf{h}^F$. The branches merge additively:

$$\mathbf{h} = \text{LN}(\mathbf{h}^T + \mathbf{h}^F). \quad (4)$$

This branch explicitly attends across feature tokens. Averaging first reduces its computational cost but removes observation order within each feature token.

An MLP maps the static vector $\mathbf{s}_n \in \mathbb{R}^9$ to γ_n and β_n , giving

$$\tilde{\mathbf{h}} = \text{LN}[\mathbf{h} \odot (1 + \gamma_n) + \beta_n]. \quad (5)$$

The final conditioning layer is zero-initialised, so the initial affine modulation is neutral. Layer normalisation still applies; the full FiLM block is not an identity map. A fusion MLP of widths 192 and 128 and a shared head of widths 256 and 128 produce classification logits and regression outputs. Fig. 3 shows the implemented data flow.

C. Optional radar branch

The CNN variant encodes two-channel Sentinel-1 VV/VH chips and adds their projection through a learned gate. For radar embedding $\mathbf{v}$, presence indicator m and image age a , the fusion is

$$\mathbf{g} = \sigma\{\text{MLP}[\tilde{\mathbf{h}} \parallel \mathbf{v} \parallel m \parallel (a/30)]\}, \quad (6)$$

$$\mathbf{h}^S = \text{LN}(\tilde{\mathbf{h}} + m \mathbf{g} \odot \mathbf{W}_v \mathbf{v}). \quad (7)$$

The gate’s final bias is initialised to -3 , giving a small, nonzero opening; absent frames contribute zero through m . A scalar variant instead appends four image-derived summaries to the dynamic stream. Experiment notes report imagery at nine of 51 nodes. Pretraining support exists in the code, but a completed pretrained checkpoint is not verified in the recorded evidence. The CNN result is therefore not presented as a validated pretrained-radar experiment.

D. Multi-task optimisation and calibrated inference

The reference objective sums masked binary cross-entropy (BCE) over four heads, with weights (1.0, 0.3, 0.3, 0.5), and adds the mean Huber loss over the two regressions with weight 0.2. The alternative `focal_conf` objective uses focal loss

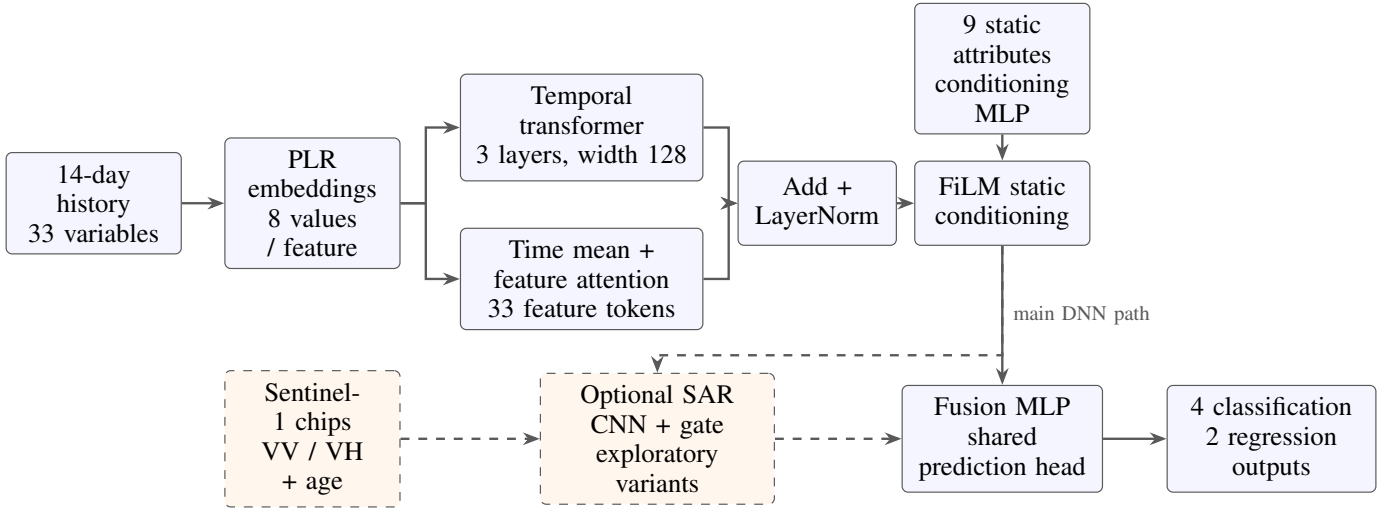


Fig. 3. MMF-Net data flow. Temporal and cross-feature attention are parallel branches sharing numerical embeddings. Static attributes condition their merged state. The principal configuration uses the solid path; optional radar fusion is applied before the fusion MLP. No graph message passing is used.

with $\alpha = 0.75$, $\gamma = 2$, and a label-confidence weight between 0.5 and 1 that decreases near the current-day discharge threshold. The comparison changes class weighting, focusing and confidence weighting jointly; it is not an isolated test of the focal exponent.

AdamW uses a learning rate of 10^{-3} , weight decay 10^{-4} , five warmup epochs and a cosine schedule. Training lasts at most 60 epochs, with gradient norm clipping at 1.0 and early-stopping patience of ten. Checkpoints are selected by validation episode-level average precision, falling back to node-day average precision if the episode score is undefined.

For five-seed ensembles, the implementation averages predicted probabilities, not logits. If p_k is the probability from seed k , calibration uses

$$\bar{p} = \frac{1}{5} \sum_{k=1}^5 p_k, \quad p_{\text{cal}} = \sigma \left[\frac{\text{logit}(\bar{p})}{T} \right]. \quad (8)$$

The temperature T is fitted on validation data. The decision threshold is selected by validation F1 and frozen for test evaluation. The reported operating points do not use a common false-alarm budget.

V. EXPERIMENTAL DESIGN

A. Main DNN ablations and historical references

The main evidence is the August 2026 MMF-Net result snapshot retained in the repository. N0–N3 successively introduce PLR embeddings, feature attention and FiLM conditioning. N4 replaces BCE with `focal_conf`; N5 adds a five-seed ensemble and temperature scaling to that objective. N5_bce applies the same ensemble procedure to N3 with BCE. N6 variants explore radar scalars and gated CNN fusion under `focal_conf`. The radar rows therefore do not continue the BCE configuration under a matched objective.

Earlier experiments include a GRU-based spatiotemporal graph network (TF-STGNN) and persistence, climatology,

TABLE I
MAIN TEMPORAL DNN CONFIGURATION.

Setting	Value
Training / validation	2003–2017 / 2018–2020
Test input dates	2021–2024
Input dimensions	14 days; 33 dynamic, 9 static
Temporal / feature blocks	3 / 1; width 128, 4 heads
Batch size	32 daily snapshots; CNN: 8
Optimiser / maximum epochs	AdamW / 60
Seeds	N0–N4: 1; N5 and N6: 5
Main target	Next-day discharge exceedance
Checkpoint / threshold	Validation episode AP / F1

discharge-percentile and LightGBM references. Those runs predate the December 2024 truncation policy. A direct ranking against MMF-Net would confound data extent with the model. These historical results appear in a separate table and are excluded from the main ablation plots.

B. Metrics and their interpretation

The stored metric named PR-AUC is computed as stepwise average precision (AP), $\sum_i p_i \Delta R_i$, rather than a trapezoidal area. We use AP throughout. The implementation sorts individual samples without grouping equal scores; tied-score references, especially binary persistence, need a standard tie-aware recomputation before direct comparison.

Node-day quantities are probability of detection, $\text{POD} = TP/(TP + FN)$; false-alarm ratio, $\text{FAR} = FP/(TP + FP)$; and critical success index, $\text{CSI} = TP/(TP + FP + FN)$. FAR is the fraction of positive predictions that are false alarms, not $FP/(FP + TN)$. Brier score averages $(p - y)^2$. Expected calibration error (ECE) uses 15 equal-width probability bins, weighted by sample counts. Under class imbalance, a small aggregate ECE can coexist with poor calibration in sparsely populated high-risk bins.

TABLE II
MAIN DNN RESULTS ON THE TEMPORAL TEST SPLIT.

ID	Configuration	AP	ECE	FAR	Params
N0	Linear embeddings	.6083	.0024	.412	.551
N1	+ PLR	.7605	.0053	.325	.555
N2	+ feature attention	.7738	.0056	.313	.693
N3	+ static FiLM	.8269	.0032	.110	.711
N4	+ focal/confidence	.7592	.0525	.138	.711
N5	N4, ensemble + TS	.7846	.0043	.145	.711
N5_bce	N3, ensemble + TS	.8355	.0016	.120	.711
N6-sc	SAR scalars, focal	.7284	.0055	.344	.719
N6-gat	SAR CNN, focal	.8310	.0066	.107	11.967

TS: temperature scaling. N0–N4: one seed; others: five. Params: millions per member. “Focal”: combined focal/confidence objective. Bold marks the largest recorded AP, not significance. Source: model-2 snapshot, 10–11 August 2026.

Episode detection credits any alarm at a node during the seven days preceding its episode onset. Validation episode AP scores each eligible episode by the maximum probability in that window and uses flood-free node-weeks as negatives. This is a retrospective association statistic. Because the model predicts the next day, an alarm six days before onset is not evidence of a calibrated six-day forecast. An onset-specific assessment should align the evaluation window with the stated horizon and require complete pre-onset coverage.

VI. RESULTS

A. Neural representation and static conditioning

Table II reports the recorded MMF-Net results. PLR embeddings increase AP from 0.6083 to 0.7605, an absolute change of 0.1522. Feature attention produces a further increase to 0.7738, followed by 0.8269 with static FiLM conditioning. These observations are consistent with the value of flexible scalar representations and location-dependent conditioning. The FiLM contrast does not isolate elevation from the drainage proxy, zone and position indicators supplied to the same module.

N0–N4 each use one seed. Their differences describe these runs rather than statistically established component effects. The reported 0.3375–0.3856 range across N5_bce seeds concerns best-validation *episode* AP. It cannot serve as an uncertainty interval for differences in test *node-day* AP. Paired test predictions and repeated controlled runs are needed for an uncertainty analysis on the same estimand.

B. Objective, ensemble and calibration

Replacing BCE with the combined focal/confidence objective changes N3’s AP from 0.8269 to 0.7592 and ECE from 0.0032 to 0.0525. In the ensemble comparison, N5_bce records AP 0.8355 and ECE 0.0016, compared with 0.7846 and 0.0043 for N5. These observations favour BCE under the evaluated settings. They do not establish that focal loss is generally unsuitable for flood prediction or identify which part of the combined weighting causes the change.

The N5_bce ensemble has reported Brier score 0.0080 and CSI 0.561. Its AP exceeds N3’s by 0.0086. Ensembling and temperature scaling are changed together, so their separate

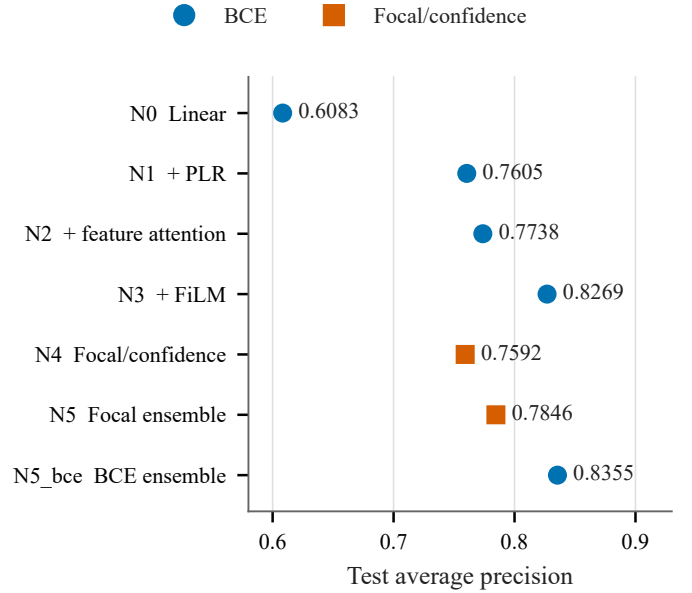


Fig. 4. Recorded DNN average precision. Points show individual configurations. Marker shape identifies the training objective. No confidence intervals are available; the plot does not establish statistical significance.

effects on probability quality are not identified. A positive temperature preserves score ordering within a fixed ensemble; it changes probabilities and threshold behaviour rather than AP. Reliability curves, bin counts and event-stratified calibration would be more informative than ECE alone, but cannot be recovered from the available aggregate results.

C. Radar fusion and model size

The scalar radar variant records AP 0.7284. The gated CNN records 0.8310, compared with 0.7846 for the image-free focal/confidence ensemble and 0.8355 for the image-free BCE ensemble. This is mixed evidence: the CNN result is higher under the focal/confidence family, yet does not exceed the strongest recorded image-free result. Differing objectives, batch sizes, uncertain pretraining status and limited coverage prevent attribution to radar alone.

Fig. 5 illustrates the resource implications. N5_bce has approximately 0.711 million reported parameters per member, whereas N6-gat has 11.967 million. A five-member image-free ensemble stores approximately 3.56 million parameters across its members and requires five forward passes. The training code counts parameters with gradients enabled, so checkpoint freeze status must accompany any final model-size comparison. Device-level latency and deployment energy were not measured.

D. Detection at recorded operating points

Fig. 6 shows event detection alongside node-day FAR for selected DNN configurations. N2 detects 0.423 of episodes under the seven-day rule at FAR 0.313; N3 records detection 0.197 at FAR 0.110; and N5_bce records 0.220 at FAR 0.120. Each point reflects both the model and its validation-selected

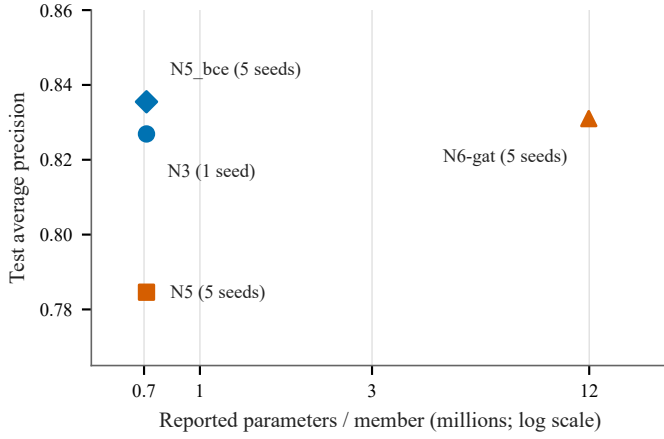


Fig. 5. Reported parameter count per member and AP for selected DNNs. The horizontal axis is logarithmic. The gated CNN has about 16.8 times the reported parameters of the image-free model. Counts do not measure latency, memory consumption or the benefit of radar information.

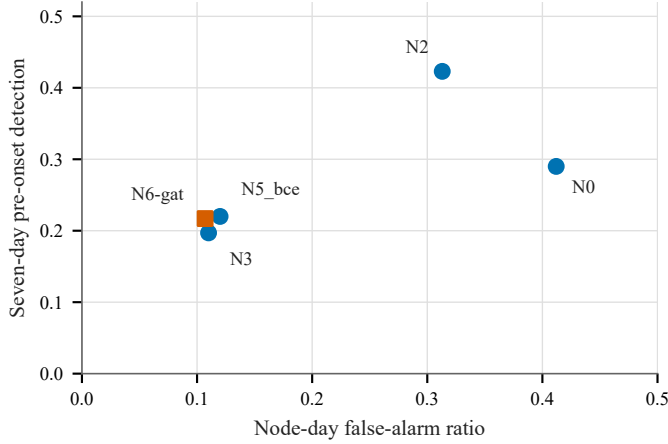


Fig. 6. Selected DNN operating points. Detection credits any alarm in the seven days before onset; FAR is calculated over node-days. Each point uses its own validation-selected threshold. The axes have different aggregation units and are not an event-level precision–recall curve.

threshold. These are not points along a shared operating curve and do not establish which network is preferable at a fixed false-alarm budget.

E. Historical references and split diagnostics

The historical discharge-percentile reference achieves AP approximately 0.816, consistent with substantial persistence in a next-day discharge target. LightGBM records 0.8496 on its historical panel (Table III). These values motivate strong references in a matched rerun; they do not establish a DNN-versus-tree ordering on an identical test set. Comparing graph-based M3 with MMF-Net would likewise change the encoder and panel as well as graph structure.

The M3 diagnostic records AP 0.704 under temporal splitting and 0.905 under random splitting, while seven-day event detection changes from 0.380 to 0.089. The random protocol interleaves node-days across train and test. Furthermore, the

TABLE III
HISTORICAL REFERENCES (2 AUGUST 2026); PANEL EXTENT DIFFERS FROM TABLE II. DASHES INDICATE UNAVAILABLE VALUES.

Reference	AP	Brier	FAR	Detection
Persistence	.590	–	.231	.223
Climatology	.108	–	.882	.515
Discharge percentile	.816	–	.231	.223
LightGBM	.8496	.0076	.080	.161
GRU only (M0)	.598	.0137	.441	.262
TF-STGNN (M3)	.704	.0135	.342	.380

event routine receives only held-out rows: random masking can remove pre-onset alarm opportunities and shift the first observed episode day. The diagnostic demonstrates sensitivity to the split, but does not isolate a causal loss of warning skill or prove memorisation of neighbouring days. The Gin holdout (AP 0.840) concerns spatial transfer on four inventory nodes, with training at other nodes spanning later dates. It is not a future-time test. Strict spatial evaluation also requires reconsidering held-out-node preprocessing statistics and graph access.

VII. PRACTICAL INTERPRETATION AND LIMITATIONS

A. From proxy prediction to warning support

The proposed use is a candidate ranking and probability component for an analyst’s river-monitoring workflow. A practical workflow would assemble available inputs at a defined daily issue time, verify freshness and coverage, run the network, and present probabilities alongside measured water levels and recent rainfall. The responsible agency would determine how outputs relate to local warning stages and actions. This is a proposed application, not a system validated by the present experiments.

The one-day target requires particular care. A daily total for day t is fully known only after that day, and reanalysis fields may be published or revised later. Operational lead time must be measured from actual input availability, not simply from date labels. Replay using archived, issue-time observations and forecasts is needed before claiming 24-hour warning capability. Radar images must similarly be restricted to acquisitions available at issue time, with image age tracked explicitly.

B. Independent evidence and failure cases

The historical validation report compares the proxy state against selected documented flood windows. Restricting its list to 2003–2024, the high-flow threshold activates in seven of ten windows; the lower threshold activates in eight. The high threshold misses the listed May 2003, June 2014 and May 2017 windows. This is a limited check of label agreement, not DNN event detection, and it does not estimate warning accuracy on a representative sample. The reported successes also use basin-level matching and expanded date windows, which are less demanding than correct daily predictions at individual nodes.

Observed gauge levels, discharge and independently mapped inundation would allow evaluation of failures that reanalysis labels cannot expose. Cases should include missed floods, false alarms, rapid-onset events and regulated rivers, alongside correct predictions. An event hydrograph should display issue time, available rainfall and discharge, model probability, the frozen threshold and independently recorded onset. Such plots require aligned observations and predictions; aggregate scores cannot supply them.

C. Temporal boundaries and reproducibility

The 14-day windows may include earlier-period features at split boundaries. Using preceding validation-period observations to initialise a test forecast is causal when those observations were available. A separate issue arises because auxiliary future targets are constructed before split assignment and input truncation. Horizons near a boundary can reach into the next block or beyond the declared input end date. A final rerun should purge boundary-crossing supervised targets and verify complete horizons explicitly.

Scores were retained from a hand-compiled experiment snapshot; underlying predictions, exact run manifests and post-truncation counts must accompany a submission. Repeated inspection of the same test set during architecture development also means these results should be treated as development evidence. A new time period or prespecified external evaluation is needed for final confirmation.

D. Priority experiments

The next DNN comparison should hold the panel, targets, seeds, optimisation budget and test rows fixed. It should repeat the embedding and FiLM ablations, separating static-attribute contributions where possible. Graph message passing can then be added to the same encoder; the unrun STG-Former implementation in the repository is not evidence of a benefit. A radar comparison should use the same BCE objective, a documented checkpoint, and results on both common-coverage nodes and the complete panel.

For decision-oriented evaluation, thresholds should be fitted on validation under a stated FAR constraint and then frozen. Realised test FAR must still be reported because the validation constraint may not hold under distribution shift. Event detection should require full pre-onset coverage and horizon-aligned alarms, supplemented by false warning episodes per basin-month. Paired uncertainty intervals should preserve storm or temporal dependence across nodes; calibration should include bin counts and high-probability subsets. These are additional experiments, not outcomes of the current study.

VIII. CONCLUSION

This study examines a terrain-conditioned deep neural network for next-day high-flow prediction at Sri Lankan river-basin locations. Recorded ablations show improvements after periodic numerical embeddings and static FiLM conditioning are introduced, while the BCE ensemble attains the highest

recorded MMF-Net AP of 0.8355. Radar results remain exploratory and the available experiments do not isolate the value of graph structure. The findings support further controlled study of neural representation and fusion. Establishing a practical warning benefit requires independent flood observations, issue-time input replay and matched evaluation of onset detection, false alarms and probability calibration.

DATA AND CODE AVAILABILITY

Code and catalogue files are available at <https://github.com/heshannethmina/Srilanka-Flood-Data-Set-Creation>. The complete training panel, checkpoints and predictions are absent from the reviewed checkout. Figure inputs and provenance accompany the manuscript.

ACKNOWLEDGMENT

We acknowledge NASA POWER, Copernicus GloFAS, Open-Meteo and Sentinel-1 data providers, public flood-report sources and Natural Earth.

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